

1 BEAM SHEAR STRESS

$$v = \frac{VQ}{IB} \quad (1)$$

v shear stress at evaluation point
 V internal shear force
 Q first moment of area about neutral axis
 I second moment of area about neutral axis
 B width at evaluation point

2 CONCRETE TYPES

TYPE	USE
I	General purpose
II	Moderate sulfate resistance; moderate heat
III	Early strength
IV	Low heat
V	High sulfate resistance

3 STANDARD PENETRATION TEST (SPT)

ITEM	NOTE
Purpose	Indirect estimate of relative density, D_r
Seat	Drive first 6 in
Count	Blows required to drive next 12 in

EXAMPLE

DRIVE	BLOWS	USE
First 6 in	5	Seating; exclude
Next 6 in	7	Include
Final 6 in	9	Include

$$N = 7 + 9 = 16 \quad (2)$$

N SPT blow count for final 12 in [blows/ft]